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| Inventor(s) | Frischen; Andreas et al. |

## Measuring Device

### Abstract

A measuring device, in particular a diagnostic device for walls, includes at least one radar sensor unit for providing radar data of a wall for which diagnostics are to be performed, a diagnostic module for performing wall diagnostics based on the radar data and for generating diagnostic results, and a display unit for displaying the diagnostic results to a user of the measuring device. The diagnostic module is configured to detect an object formed in the wall based on the radar data, wherein the object detection comprises a determination of an object position and a classification of an object type for the object.

**Inventors:** **Frischen; Andreas (Leonberg, DE), Koelsch; Andreas (Peiting, DE), Giliard; Anke (Stuttgart, DE), Moster; Benjamin (Niefern-Oeschelbronn, DE), Sgarz; Heiko (Leonberg, DE), Steuler; Jan Linus (Woelfersheim, DE), Fiederer; Lukas (Balingen, DE), Troeltzsch; Uwe (Roesrath, DE)**

**Applicant:** **Robert Bosch GmbH** (Stuttgart, DE)

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## Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 202 230.6, filed on Mar. 11, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.
[0002] The present disclosure relates to a measuring device, in particular to a wall diagnostic device.
BACKGROUND
[0003] Wall diagnostic devices for performing wall diagnostics and detecting objects formed in the walls are known from prior art.
[0004] The present disclosure provides an improved measuring device, in particular a wall diagnostic device.
SUMMARY
[0005] According to one aspect, a measuring device is provided, in particular a diagnostic device for walls, comprising at least one radar sensor unit for providing radar data of a wall for which diagnostics are to be performed, a diagnostic module for performing wall diagnostics based on the radar data and for generating diagnostic results, and a display unit for displaying the diagnostic results to a user of the measuring device, wherein the diagnostic module is configured to detect an object formed in the wall based on the radar data, wherein the object detection comprises a determination of an object position and a classification of an object type for the object.
[0006] This thus achieves the technical advantage that an improved measuring device can be provided, in particular a wall diagnostic device. The measuring device is preferably configured as a wall diagnostic device, by means of which walls to be examined can be examined. In particular, the wall diagnostic device is configured to detect objects located in the wall. For this purpose, the measuring device comprises at least one radar sensor unit, by means of which radar signals can be transmitted towards a wall to be examined and radar signals reflected at the front of the wall can be received. Furthermore, the measuring device comprises a diagnostic module configured to execute on the radar data of the radar sensor unit and to perform wall diagnostics based thereon.
[0007] The diagnostic module is configured to perform object detection of objects arranged in the wall to be examined based on the radar data of the radar sensor unit. The object recognition comprises at least one object detection and one object classification.
[0008] The object detection comprises detecting an object located in the wall and determining an object position of the object within the wall. The object classification comprises determining object classes, with which the detected object is to be associated. With the object classification, object types of the detected object may be unambiguously determined. In addition, the measuring device comprises a display unit. On the display unit, which may be configured as a display, for example, the diagnostic results of the wall diagnostics, i.e., an object position and/or an object type of the detected and classified object located in the wall, can be displayed to a user.
[0009] The measuring device according to the disclosure has the advantage that an unambiguous classification of the detected objects, i.e., a clear association of object types with the detected objects, can be carried out exclusively based on radar data of a radar sensor unit of the measuring device. Additional information, for example measured values of additional measurement variables of additional sensor elements, is not necessary for object recognition in the sense of the present disclosure.
[0010] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.
[0011] The technical advantage can thereby be achieved that, based on the radar data of the radar

sensor unit, an object depth of the object within the wall can be determined by the measuring device in addition to the object recognition. The object depth describes a distance of the object to a surface of the wall. The correspondingly configured diagnostic module may determine the object depth based solely on the radar data of the radar sensor unit.

[0012] According to one embodiment, the diagnostic module is further configured to determine an object extension of the object in a predefined direction based on the radar data.

[0013] The technical advantage can thereby be achieved that an object extension of the object detected in the wall is further enabled based on the radar data of the radar sensor unit. The object extension describes an extension of the object at least with respect to one spatial direction, preferably with respect to two spatial directions, especially preferably with respect to three spatial directions. This allows for one-dimensional, preferably two-dimensional, more preferably three-dimensional extension information for the object located in the wall.

[0014] According to one embodiment, the diagnostic module is further configured to classify a wall type of the wall based on the radar data.

[0015] The technical advantage can thereby be achieved that, in addition to object recognition, wall type classification can also be performed based solely on the radar data from the radar sensor unit. In the wall type classification, various wall types of the walls to be examined can be determined or classified by the correspondingly configured diagnostic module. Due to the fact that the wall type is classified automatically, the user does not have to enter the respective wall type of the wall to be examined as additional information for the wall diagnostics, as is known from prior art. Instead, the wall type is automatically determined by the diagnostic module based on the radar data from the radar sensor unit.

[0016] The wall type may be incorporated into the wall diagnostics and object recognition as additional information, for example for background correction. Alternatively or additionally, the detected wall type may be displayed to the user in the display unit as additional information in the diagnostic result.

[0017] According to one embodiment, the diagnostic module is further configured to perform a subsurface correction for object recognition based on the wall type classification.

[0018] The technical advantage can thereby be achieved that more precise object detection may be realized by considering the wall type classification of the wall to be examined in a subsurface correction of the radar data of the radar sensor unit. Different wall types provide different radar signals reflected by the wall. When taking the respective wall type into consideration in the form of a subsurface correction, these effects of the different wall types on the radar signals can be taken into account. This allows for false object recognition due to the wall type to be avoided and allows for more precise detection of the objects located in the wall.

[0019] According to one embodiment, the display unit is configured to display to the user the object position of the object in the wall and a type class of the object type of the object and/or an object depth and/or an object extension of the object and/or a type class of the wall type of the wall as a diagnostic result.

[0020] The technical advantage can thereby be achieved that a variety of different information regarding the wall to be examined may be provided to the user in the display of the measuring device. The user thus has the essential results of the wall diagnostics at a glance in the display unit and can carry out the planned work on the wall based on this.

[0021] According to one embodiment, the diagnostic module comprises at least a pre-processing module, a wall type classification module and an object recognition module, wherein the pre-processing module is configured to pre-process the radar data of the radar sensor unit and provide input data for the wall type classification module and the object recognition module, wherein the wall type classification module is configured to classify the wall type of the wall based on the input data provided by the pre-processing module, and wherein the object recognition module is configured to detect the object formed in the wall and classify the respective type of object of the object based on the input data provided by the pre-processing module.

[0022] The technical advantage can thereby be achieved that precise wall diagnostics are enabled by the correspondingly configured diagnostic module using the radar data of the radar sensor unit. By pre-processing the radar data of the radar sensor unit, the radar data can be brought into the form required for wall diagnostics. Using a correspondingly configured wall type classification module, a corresponding wall type classification can be performed on the pre-processed radar data, and the respective wall type of the wall can be determined.

[0023] An object recognition of an object located in a wall may be performed by an object recognition module based on the pre-processed radar data and taking into account the wall type provided by the wall type classification module. The object recognition module is configured to perform an object detection and object classification of the object located in the wall based on the pre-processed radar data and taking into account the provided wall type. The presented architecture of the diagnostic module allows for the most precise and reliable wall diagnostics of walls to be examined.

[0024] According to one embodiment, the diagnostic module comprises at least a first pre-processing module and a second pre-processing module, wherein the first pre-processing module is configured to pre-process the radar data of the radar sensor unit and provide the input data for the wall type classification module, wherein the wall type classification module is configured to classify the wall type of the wall based on the input data provided by the first pre-processing module and provide wall type information to the second pre-processing module, wherein the second pre-processing module is configured to pre-process the radar data of the radar sensor unit and, taking into account the wall-type information of the wall-type classification module, provide the input data to the object recognition module, and wherein the object recognition module is configured to detect the object formed in the wall and classify the respective type of object of the object based on the input data provided by the second pre-processing module.

[0025] The technical advantage can thereby be achieved that the first pre-processing module and the second pre-processing module enable more precise pre-processing of the radar data of the radar sensor unit. The presented architecture of the diagnostic module thus enables further precise object recognition and thus further precise wall diagnostics.

[0026] According to one embodiment, the measuring device further comprises at least one of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or a nuclear magnetic resonance (NMR) sensor and/or an ultrasonic sensor for providing additional sensor data, wherein the diagnostic module is configured to perform the wall diagnostics by taking into account the additional sensor data.

[0027] The technical advantage can thereby be achieved that, through further sensor data, additional sensors, each of which is configured to detect different physical variables, can provide additional information in addition to the radar data of the radar sensor unit, for incorporation into the wall diagnostics. This additional information, which is preferably complementary to the information of the radar data of the radar sensor unit, allows further precise specification of the wall diagnostics and the object recognition, respectively.

[0028] According to one embodiment, the diagnostic module comprises at least one correspondingly trained artificial intelligence structure configured to perform an object recognition and/or wall classification and/or object depth determination based on the radar data and/or the additional sensor data.

[0029] The technical advantage can thereby be achieved that, because the diagnostic module is configured as a correspondingly trained artificial intelligence structure, which is trained based on the radar data, or, if applicable, taking into account the information of the additional sensors, to perform an object recognition and/or a wall classification and/or an object depth determination and/or an object extension determination, a reliable and powerful diagnostic module can be provided. By using the artificial intelligence technology, precise wall diagnostics can be provided.

[0030] According to one embodiment, the measuring device further comprises a motion detection unit, wherein the motion detection unit is configured to sense movement of the measuring device along the surface of the wall.
[0031] The technical advantage can thereby be achieved that a movement of the measuring device relative to the wall can be determined by the motion detection unit. In the customary use of wall diagnostic devices, the wall diagnostic device is moved by the user along the length of the wall, on which the diagnostics are performed. The corresponding relative movement of the measuring device relative to the wall can be determined by the motion detection unit. Based on this determined relative movement, the wall diagnostics may be performed for different positions of the measuring device relative to the wall. This allows a surface examination of the wall to be examined and allows the recognition of objects and the determination of extension of objects over a range that is substantially larger than the effective range of the radar sensor unit.
[0032] The motion detection unit thus allows wall diagnostics while the measuring device is moving relative to the wall, as a result of which a larger area of the wall to be examined can be covered and the wall diagnostics can be accelerated accordingly.
[0033] According to one embodiment, object classes of the detected object comprise: metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, and/or wherein the wall type classes of the wall comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, underfloor heating, wall heating.
[0034] The technical advantage may thereby be achieved that a large number of different objects of different object types can be detected or classified. The measuring device or the diagnostic module can be trained hereby to detect and classify common objects installed in building walls. This allows particularly precise wall diagnostics, in which the detected objects can be precisely and unambiguously assigned to the corresponding object classes.
[0035] By precisely classifying objects and providing the corresponding classification information to the user via the display unit, wall diagnostics are made as meaningful as possible. Since the user not only knows that an object is located within the wall and where it is, but also which object type the detected object is, the user can decide accordingly how to carry out further work on the wall with respect to the detected object. Providing the object types of the object classification of the detected objects thus represents an essential area of the wall diagnostics, because the user can adjust the planned work on the wall based on the indicated object type accordingly.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0036] Embodiments of the disclosure are described with reference to the following figures. The figures show:
[0037] FIG. **1** a schematic illustration of a measuring device according to one embodiment;
[0038] FIG. **2** a further schematic illustration of the measuring device according to a further embodiment;
[0039] FIG. **3** a further schematic illustration of the measuring device according to a further embodiment;
[0040] FIG. **4** a schematic illustration of a measurement of the measuring device according to one embodiment, and
[0041] FIG. **5** a further schematic illustration of the measuring device according to a further embodiment.
DETAILED DESCRIPTION

[0042] FIG. **1** shows a schematic illustration of a measuring device **100** according to one embodiment.
[0043] The present disclosure relates to a measuring device, in particular a wall diagnostic device for examining walls **105** to be worked on. Wall diagnostic devices are known from prior art that are used to detect objects located in walls. Such devices allow a user to examine walls to be worked on to search for objects located in the walls, in order to be able to perform planned work, for example drilling in walls, such that damage to the objects located in the walls can be avoided.
[0044] In the embodiment shown, the measuring device **100** comprises a housing **150** having a handle **152** for grasping of the measuring device **100** by a user, a display unit **111** for displaying diagnostic results **109** of the wall diagnostics, and controls **154** for switching the measuring device **100** between various operating modes.
[0045] According to the disclosure, the measuring device **100** comprises at least one radar sensor unit **101**. By means of the radar sensor unit **101**, radar signals may be transmitted towards the wall **105** to be examined, and radar signals reflected by the wall **105** may be received.
[0046] For example, the radar sensor unit **101** may be configured as a narrow band radar detector device in the 2.4 GHz to 2.4835 GHz frequency range or as an ultra-wide band radar detector device in the 1.8 GHz to 5.8 GHz frequency range.
[0047] The measuring device **100** further comprises a diagnostic module **107** executable on a computing unit **151** of the measuring device **100** to perform the wall diagnostics. The diagnostic module **107** is configured to perform corresponding diagnostics on the wall to be examined based on the radar data **103** of the radar sensor unit **101**. The radar data **103** of the radar sensor unit **101** thereby depicts the wall **105** to be examined and, if applicable, objects **113** located within the wall **105**.
[0048] The wall diagnostics carried out by the diagnostic module **107** comprise at least performing object recognition. The object recognition comprises an object detection and an object classification of the object **113** located in the wall **105**. The object detection comprises at least the determination of an object position **115**. The object position describes the positioning of the object located in the wall **105** with respect to a reference system defined by the measuring device **100**. The object classification of the detected object **113** comprises at least determining an object type **117** of the detected object **113**.
[0049] The diagnostic results of the wall diagnostics determined in this way, i.e. at least the determined object position **115** and/or the determined object type **117** of the object **113** located in the wall **105**, are subsequently presented in a display unit **111** of the measuring device **100** to a user of the measuring device **100**. For example, the display unit **111** may be configured as a corresponding display, and the diagnostic results **109** may be visually displayed. Additionally, the display of the diagnostic results **109** may be supported via audible and/or haptic signals. For example, the haptic signals may be realized via corresponding vibration signals.
[0050] The object **113** can be shown in the display, for example, by means of a corresponding icon. The object **113** can be shown in the corresponding object position **115** in the display. The object extension **121** may be visualized by a corresponding size of the displayed icon. The particular object type **117** of the object **113** may be visualized with a corresponding term or color highlighting of the icon, or by a specific shape of the icon representing the object **113**.
[0051] Alternatively, the wall diagnostics may additionally comprise determining a wall type **123** in the form of a wall type classification of the wall **105** to be examined. The wall type **123** describes the respective type of the wall **105** to be examined. For example, the wall type may be associated with corresponding wall type classes, which may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or wall bricks, underfloor heating, wall heating or similar wall types found in buildings.
[0052] According to one embodiment, the diagnostic module **107** is further configured to determine an object depth **119** of the object **113** within the wall **105** based on the radar data **103**. The object

depth **119** is defined by a distance of the object formed in the wall **105** to a surface of the wall **105**. The distance may be defined on the object side, for example with respect to an object surface or with respect to an object center point. The distance to the surface of the wall **105** describes a shortest distance defined by a direction perpendicular to the surface of the wall **105**.

[0053] According to one embodiment, the diagnostic module **107** is further configured to determine an object extension **121** of the object **113** in at least one predefined direction based on the radar data **103**. The object extension **121** of the object **113** describes a spatial extension of the object **113** in at least one spatial direction, preferably in two spatial directions, particularly preferably in three spatial directions. The object **113** may be described here as a one-dimensional, two-dimensional, or three-dimensional object **113**.

[0054] In conventional use, the measuring device **100** is placed on the surface of the wall **105** to be examined. Radar signals are transmitted towards the wall **105**, and radar signals reflected from the wall **105** or the objects **113** located behind it are received, via the radar sensor unit **101**. This radar data **103** of the radar sensor unit **101** is used by the diagnostic module **107** to perform the wall diagnostics described above and to determine corresponding diagnostic results **109**.

[0055] For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** of the object **113** located in the wall **105**. Alternatively or additionally, the diagnostic results **109** may comprise the wall type **123** of the wall **105** and/or the object depth **119** and/or the object extension **121** of the object **113**.

[0056] The diagnostic results **109** configured in this manner may subsequently be displayed to a user of the measuring device **100** in a display unit **111** of the measuring device **100**. The display unit **111** can be configured as a corresponding display, for example. The diagnostic results **109** may be displayed in graphical or textual form in the display unit **111**.

[0057] According to one embodiment, the measuring device **100** further comprises a motion detection unit **141**. The motion detection unit **141** may be used to detect movement of the measuring device **100** relative to the wall **105**. The motion detection unit **141** may comprise, for example, at least one roller element for this purpose. When the roller element is placed on the wall surface of the wall **105**, movement of the measuring device **100** relative to the wall **105** can be detected when the measuring device **100** moves along a direction of movement **153** by rolling the roller element. Alternatively, the motion detection unit **141** may have a different configuration by which a relative movement of the measuring device **100** relative to the wall **105** can be detected.

[0058] By moving the measuring device **100** relative to the wall **105**, radar data **103** of the radar sensor unit **101** may be captured for a plurality of different positions of the measuring device **100** relative to the wall **105**. This allows a larger spatial area of the wall **105** to be examined than the effective range of the radar sensor unit **101**. This allows for objects **113** to be captured that have a greater spatial extent than the effective range of the radar sensor unit **101**.

[0059] While the measuring device **100** moves along the direction of movement **153**, radar data **103** of the radar sensor unit **101** may be captured continuously. The wall diagnostics may be evaluated by the diagnostic module **107** based on this radar data **103** while the measuring device **100** is moving along the direction of movement **153**. This allows for accelerated wall diagnostics, taking into account the positioning of the measuring device **100** relative to the wall **105**.

[0060] According to its embodiment, the diagnostic module **107** is configured as a correspondingly trained artificial intelligence **125**. The artificial intelligence **125** is trained to perform the above-mentioned wall diagnostics based on the radar data **103** of the radar sensor unit **101** and to determine at least the object position **115** and the object type **117** of an object **113** located in the wall **105**. The object classification or determination of the object type **117**, respectively, comprises assigning the detected object **113** to predefined object classes.

[0061] The object classes may comprise: Metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, or other elements commonly installed in building walls.

[0062] Furthermore, the artificial intelligence **125** may be trained to determine the wall type **123** of the wall **105** to be examined at least based on the radar data **103** of the radar sensor unit **101**. Possible wall types **123** may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or individual bricks of the brick wall, underfloor heating, wall heating or other similar wall types commonly installed in buildings.

[0063] According to one embodiment, in addition to the radar sensor unit **101**, the measuring device **100** may comprise further additional sensors by means of which additional physical variables are detectable. For example, the measuring device **100** may comprise an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor or other sensors commonly used in wall diagnostic devices.

[0064] The diagnostic module **107**, in particular the corresponding trained artificial intelligence **125**, can be configured to perform wall diagnostics based on the radar data **103** of the radar sensor unit **101** and taking into account the additional sensor information of the further sensors described above. The additional information of the additional sensors mentioned above can particularly be used for object recognition of the objects **113** located in the walls **105**. The additional sensor information may possibly provide improved detection of the objects **113** and may possibly provide improved classification of the objects **113**.

[0065] For example, the material of the objects **113**, for example as a metallic or non-metallic material, can be particularly improved and classified by using the additional sensor information.

[0066] FIG. **2** shows another schematic representation of the measuring device **100** according to a further embodiment.

[0067] In the embodiment shown, the measuring device **100** comprises a pre-processing module **127** in addition to the diagnostic module **107**. For wall diagnostics, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. Pre-processing of the receiving radar data **103** is performed via the pre-processing module **127**. For example, via pre-processing by the pre-processing module **127**, the radar data may be brought into a corresponding data structure required for wall diagnostics by the diagnostic module **107**.

[0068] As described above, during wall diagnostics, the diagnostic module **107** generates the diagnostic results **109** described above. For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** and/or object depth **119** and/or object extension **121** of an object **113** formed in the wall **105** to be examined and/or the wall type **123** of the wall **105** to be examined. The correspondingly generated diagnostic results **109** may subsequently be displayed in the display unit **111** of the measuring device **100**.

[0069] According to one embodiment, in addition to the radar data **103** of the radar sensor unit **101**, the additional sensor information of the additional sensors described above may be considered during the wall diagnostics of diagnostic module **107**. A corresponding pre-processing of the additional sensor information may be performed accordingly by the pre-processing module **127**.

[0070] In the embodiment shown, the diagnostic module **107** comprises a wall type classification module **129** and an object recognition module **131**. The pre-processing module **127** comprises a first pre-processing module **135** and a second pre-processing module **137**. The first pre-processing module **135** comprises an S matrix reduction **155**. The second pre-processing module **137** comprises a background correction **157**, an inverse Fast Fourier transformation **159**, and a focusing and migration **161**. During the pre-processing of the radar data **103** by the pre-processing module **127**, the radar data **103** is first pre-processed by the first pre-processing module **135** and the S-matrix reduction **155** contained therein.

[0071] When doing so, the first pre-processing module **135** generates input data **133** based on the radar data **103**. The input data **133** serves as input data for the wall type classification module **129**. The wall type classification module **129** performs a wall type classification of the wall **105** to be examined based on the input data **133** and generates wall type information **139**. The wall type information **139** contains the wall type **123** of the wall **105** to be examined, as determined in the wall type classification.

[0072] Subsequently, the second pre-processing module **137** performs pre-processing based on the radar data **103** and the wall type information **139**. A background correction **157** of the radar data **103** is performed during this, taking into account the wall type **123** determined in the wall type information **139**. Depending on the wall type **123** of the wall **105** to be examined, different effects can occur on the radar data **103**.

[0073] These effects, which are primarily based on the respective wall type **123** and can affect object recognition, can be corrected by the background correction **157**. After the background correction has been performed, further pre-processing can be carried out by performing the inverse Fast Fourier transformation **159** or focusing and migration **161**, respectively, and input data **133** can be created for the object recognition module **131** once again. Based on the input data **133** provided by the second pre-processing module **137**, the object recognition module **133** performs object recognition of the object **113** located in the wall **105** to be examined and determines at least the object position **115** and the object type **117** of the respective object **113**. Additionally, the object depth **119** and the object extension **121** may be determined by the object recognition module **131**.

[0074] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.

[0075] The pre-processing is optional. Depending on the algorithm used for the diagnostic module **107**, completely unprocessed radar echoes of different frequencies may be used as radar data **103** and as input data for the diagnostic module **107**. Alternatively, radar data **103** processed via multiple steps may be used. The pre-processing steps comprise, for example, the transformation of the signals from the frequency domain to the time or distance domain, background deduction, noise removal, and normalization of the signals. For radar data **103** which is available in the form of complex numbers, only the absolute value can be processed. Alternatively or additionally, the phase information may be considered.

[0076] FIG. **3** shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0077] In the embodiment shown, the diagnostic module **107** comprises a plurality of parallel processing paths **102**. Each processing path **102** comprises a pre-processing module **127**, the diagnostic module **107**, comprising the wall type classification module **129** and/or the object recognition module **131**, for example, according to the embodiment in FIG. **2**, and a post-processing module **163**.

[0078] In FIG. **3**, primarily the radar data **103** is shown as input data for the wall diagnostics. In addition to the radar data shown, however, the additional information from the additional sensors may also serve as input data for the wall diagnostics. The different information from the different types of sensors in the different parallel processing paths **102** can be processed and the corresponding wall diagnostics performed separately on the different sensor information. Upon completion of the wall diagnostics, a summary module may be used to assemble a summary of the individual partial analysis results for the diagnostic results **109** of the wall diagnostics.

[0079] Alternatively or additionally, different sub-aspects of wall diagnostics may be performed by the different processing paths **102** based on the same sensor information.

[0080] For example, the individual processing paths **102** may process different radar data **103** captured during movement of the measuring device **100** relative to the wall **105** for different positions of the measuring device **100** relative to the wall **105**. The radar data **103**, thus representing different regions of the wall **105** and captured sequentially in time as the measuring device **100** moves relative to the wall **105**, may then be processed in the different processing paths **102** by the modules shown.

[0081] The various processing paths perform stand-alone wall diagnostics including at least determining the object position **115** and/or the object type **117** of the object **113** located in the wall **105**.
[0082] By means of the summary module **165**, the partial results of the stand-alone wall diagnostics of the different regions of the wall **105** provided in the individual processing paths **102** can be summarized to a contiguous diagnostic result **109**. The contiguous diagnostic result describes the wall diagnostics of a contiguous spatial area that was scanned during movement of the measuring device **100** relative to the wall **105** and depicted by the corresponding captured radar data **103**. The parallel processing of the radar data **103** or additional sensor information **104** of the additional sensor elements in the different processing paths **102** thus enables accelerated wall diagnostics.
[0083] Alternatively, various wall diagnostic functions may also be performed in the different processing paths **102**. For example, in one processing path **102**, the wall type classification and the determination of the wall type **123** of the wall **105** to be examined can be performed. In another processing path **102**, object recognition of the object **113** located in the wall can be performed. The object detection can be performed with the determination of the object position **115** and the object classification can be performed with the determination of the object type **113** in one processing path **102**.
[0084] Alternatively, the object detection and object classification may then also be performed in two separate processing paths **102**. In further processing paths **102**, the object depth determination, i.e., the determination of the object depth **119** and/or the determination of the object extension **121** may each be carried out. In the summary module **165**, the various partial results of the wall diagnostics may be summarized into corresponding diagnostic results **109**.
[0085] The diagnostic module **107** can be divided into different artificial intelligence structures **125**, as already shown in the embodiment in FIG. **2**. For example, the diagnostic module **107** may comprise a wall type classification module **129** and an object recognition module **131**. The object recognition module may in turn be divided into an object detection module and an object classification module. The diagnostic module **107** may further comprise an object depth determination module and object extension module, respectively configured to determine the object depth **119** and the object extension **121**.
[0086] The respective modules may each be configured as stand-alone artificial intelligence structures **125**, for example, neural networks. Alternatively, the various modules may form portions of an overall artificial neural network that are connected to an overall neural network according to structures known from prior art.
[0087] FIG. **4** shows a schematic illustration of a measurement by the measuring device **100** according to one embodiment.
[0088] For pre-processing, the radar data **103** or the additional sensor information **104** of the remaining sensors may be normalized, in particular to numerically stabilize the subsequent steps performed by the diagnostic module **107** during the wall diagnostics. For example, an amplitude and/or offset compensation may be performed for this purpose. Moreover, to reduce interference, filtering of the radar data **103** may be carried out, and to reduce the data rate, the corresponding sensor data may be sampled. Additionally, the radar data **103** or the additional sensor information **104** may be transformed into the respectively required frequency range or time range. Methods known from prior art can be used for this purpose.
[0089] Furthermore, the captured radar data **103** or additional sensor information **104** may be divided into temporal or spatial windows **167**. Temporal windows **167** may be generated by recording the radar data **103** or the additional sensor information or the pre-processed radar data **103** over a fixed time interval. Spatial windows **167**, on the other hand, may be generated from a mapping of the radar data **103** or additional sensor information **104** to positions of the measuring device **100** relative to the wall **105** along the direction of movement **153**.

[0090] Graph a) of FIG. **4** shows such a data matrix resulting from the steps described above. The data matrix of the window **167** shown in graph a) shows a plurality of sensor data, which may comprise, for example, radar data **103** or additional sensor information **104** of the further sensors plotted along a frequency channel axis **171** or along a space/time axis **169**, respectively.
[0091] A width of the temporal windows **167** may be selected such that different sampling rates of the sensors may be balanced and a new window **167** may be provided frequently enough so that a display of the diagnostic results **109** of the wall diagnostics in the display unit **111** may be shown without too great of a time delay while the measurement is being performed or shortly after the measurement by the measuring device **100** ends.
[0092] A rate of 2 to 20 windows per second of the data recording of the sensor data may be advantageous for this purpose. For spatial windows, the spatial sampling rates can be selected to achieve the desired local accuracy. Advantageously, 1 mm to 1 cm sampling rates can be used. This means that corresponding sensor data is captured every 1 mm to 1 cm of movement of the measuring device **100** along the direction of movement **153**.
[0093] A width of the spatial windows **167** may be selected such that contiguous information relating to an object **113** is contained in a window. Advantageously, a width of the respective spatial windows **167** can be 1 cm to 20 cm. This results in 4 to 100 measured values per window **167**. This allows for further efficient algorithmic processing of the correspondingly recorded radar data **103** or additional sensor information by the diagnostic module **107**.
[0094] A further temporal window **167** or spatial window **167** can be provided as soon as one or more sampling points are available.
[0095] The diagnostic module **107** may be oriented such that a matrix corresponding to the window size of the respective spatial or temporal window **167** may be included as input data, for example of each processing path **102** of the embodiment in FIG. **3** as well. According to the embodiment in FIG. **2**, the corresponding input data can comprise the respective pre-processed sensor data, i.e. radar data **103** and additional sensor information **104** of the additional sensors.
[0096] As stated above, the wall diagnostics may be performed by the diagnostic module **107** based on a correspondingly trained artificial intelligence. Alternatively, different processing paths may also be calculated by means of rule-based algorithms. In addition, within a processing path **102**, a combination of artificial intelligence and rule-based algorithm is possible in the form of a parallel operation or concatenation.
[0097] The diagnostic results **109** of the wall diagnostics may be expressed as numeric values, vectors, or matrices. Furthermore, for the object detection, the probability of detection, or for the wall type classification and/or the object classification, respectively, a probability of the specified object classes and/or wall type classifications can be indicated. The same may apply to the position and/or depth determination, for which corresponding probability values can also be indicated.
[0098] In addition to the radar data **103**, if the additional sensor information of the further sensor types is processed in a processing path **102**, these may either be merged within the artificial intelligence **125** or combined by means of rule-based combinations.
[0099] During the post-processing of each processing path **102**, of the embodiment in FIG. **3**, multiple algorithm results based on multiple windows **167** may be summarized by the summary module **165**. This summary can in particular be realized by means of majority formation, sum formation or also by multiplication of successive probability values.
[0100] Furthermore, by clustering multiple results, for example, it is possible to detect which objects, of multiple detected objects located close to one another, are the same object so that they are not incorrectly detected multiple times.
[0101] Likewise, it is possible to multiplicatively apply a weighting function **177** when summarizing the results from multiple windows **167**. Advantageously, the diagnostic partial results **175** corresponding to corresponding data points in the space may be weighted with respect to positioning of the diagnostic partial results **175** relative to a center point of the respective window

**167**. This is illustrated by way of example in graph b), in which the individual diagnostic partial results **175** are weighted according to the weighting function **177** shown with respect to the center point of the window **167** shown.

[0102] According to one embodiment, the results of one processing path **102** after post-processing **163** may influence the extension of another processing path **102**. Weighting parameters may be adjusted that may depend on the particular result from the processing path **102** for each window.

[0103] For example, the result of an object classification in which the object type **117** of an object located in the wall **105** is defined, can be used to increase the weight of a wall type classification, in which the wall type **123** of the respective wall **105** is determined, during the post-processing at locations without objects **113**, because the respective radar data **103** at these locations is less influenced by reflections of the objects **113**.

[0104] FIG. **5** shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0105] Graphs a) and b) of FIG. **5** show two different alternatives of joint data processing of radar data **103** and additional sensor information **104** by the diagnostic module **107**.

[0106] Graph b) illustrates a joint processing of the radar data **103** and the additional sensor information **104** of the additional sensors by the diagnostic module **107**. For this purpose, the radar data **103** and the additional sensor information **104** are collectively used as input data of the diagnostic module **107** configured as an artificial intelligence, in particular as an artificial neural network. The diagnostic module **107** here comprises multiple convolutions **108** and multiple dense layers **106**. The radar data **103** and the additional sensor information **104** are processed jointly as input data via the convolutions **108** and dense layers **106**. The aforementioned diagnostic results **109** are created as output data of the diagnostic module **107** based on this.

[0107] On the other hand, in graph b) the radar data **103** and the additional sensor information **104** are used as stand-alone input data of the diagnostic module **107**. The diagnostic module **107** becomes multiple processing paths **102**. The processing paths **102** each comprise multiple convolutions **108** and at least one dense layer **106**. In the various processing paths **102**, wall diagnostics are performed separately by the diagnostic module **107** based on the radar data **103** and the additional sensor information **104**, respectively.

[0108] In an additional concatenation layer **148**, the partial results of the partial diagnostics of the different processing paths **102** are combined and fed to a final dense layer **106**. The output data of the diagnostic module **107** corresponds to the diagnostic results **109** described above.

[0109] The correspondingly configured diagnostic module **107** is configured to perform wall diagnostics as described above, including the features described above, based on the radar data **103** and the additional sensor information **104**.

[0110] In the embodiment shown, the diagnostic module **107** is configured as an artificial neural network, in particular as a convolutional network. Corresponding network architectures with convolutions **108**, dense layers **106**, and concatenation layers **148** are known from prior art.

## Claims

**1**. A measuring device, comprising: at least one radar sensor unit configured to provide radar data of a wall for which diagnostics are to be performed; a diagnostic module configured to perform wall diagnostics based on the radar data and to generate diagnostic results; and a display unit configured to display the diagnostic results to a user of the measuring device, wherein the diagnostic module is configured to detect an object formed in the wall based on the radar data, and wherein the detecting of the object comprises a determination of an object position and a classification of an object type for the object.

**2**. The measuring device according to claim 1, wherein the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, the object depth being defined by a distance of the object formed in the wall to a surface of the wall.

**3**. The measuring device according to claim 1, wherein the diagnostic module is further configured to determine an object extension of the object in a predefined direction based on the radar data.

**4**. The measuring device according to claim 1, wherein the diagnostic module is further configured to classify a wall type of the wall based on the radar data.

**5**. The measuring device according to claim 4, wherein the diagnostic module is further configured to perform a subsurface correction for object recognition based on the classification of the wall type.

**6**. The measuring device according to claim 1, wherein the display unit is configured to display to the user the object position of the object in the wall and at least one of a type class of the object type of the object, an object depth, an object extension of the object, and a type class of a wall type of the wall as the diagnostic results.

**7**. The measuring device according to claim 4, the diagnostic module comprising: at least one pre-processing module configured to pre-process the radar data of the radar sensor unit and provide input data; a wall type classification module configured to classify the wall type of the wall based on the input data provided by the at least one pre-processing module; and an object recognition module configured to detect the object formed in the wall and to classify the respective object type of the object based on the input data provided by the at least one pre-processing module.

**8**. The measuring device according to claim 7, wherein: the at least one pre-processing module comprises a first pre-processing module and a second pre-processing module, wherein the first pre-processing module is configured to pre-process the radar data of the radar sensor unit and provide the input data for the wall type classification module, wherein the wall type classification module is configured to classify the wall type of the wall based on the input data provided by the first pre-processing module and provide wall type information to the second pre-processing module, wherein the second pre-processing module is configured to pre-process the radar data of the radar sensor unit and, taking into account the wall type information of the wall type classification module, provide the input data to the object recognition module, and wherein the object recognition module is configured to detect the object formed in the wall and classify the respective object type of the object based on the input data provided by the second pre-processing module.

**9**. The measuring device according to claim 1, further comprising: at least one of an induction sensor, an eddy current sensor, a capacitance sensor, an AC current sensor, a nuclear magnetic resonance (NMR) sensor, and an ultrasonic sensor configured to provide additional sensor data, wherein the diagnostic module is configured to perform the wall diagnostics by taking into account the additional sensor data.

**10**. The measuring device according to claim 1, wherein the diagnostic module further comprises at least one correspondingly trained artificial intelligence configured to perform object recognition and/or a wall classification and/or an object depth determination based on the radar data and/or additional sensor data.

**11**. The measuring device according to claim 1, further comprising: a motion detection unit configured to sense a movement of the measuring device along a surface of the wall.

**12**. The measuring device according to claim 1, wherein: the classification of the object type includes classifying the object as at least one selected from the group consisting of: metallic or non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, fresh water pipe, non-water filled plastic pipe, and waste water pipe.

**13**. The measuring device according to claim 6, wherein the wall type classification module is configured to classify the wall type as one or more selected from the group consisting of: concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, underfloor heating, and wall heating.